

Sex Differences in the Neurobiology of Stress



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KEYWORDS

• Stress • Sex differences • Neurobiology • Estrogen • Stress-mitigating interventions

KEY POINTS

- Women disproportionately experience lasting physical effects of stress when compared with their male counterparts.
- Chronic stress can increase HPA axis dysregulation and disrupt glucocorticoid balance in women, which negatively impact neurobiology and cognition.
- Differences in neural plasticity fueled by hormonal differences between women and men could contribute to increased susceptibility of women to stress.

INTRODUCTION

When a person perceives a stressor, their body undergoes distinct physical reactions to respond to the stressor. Allostasis refers to the body's coordinated response to a stressor to respond to the challenge and bring organ systems back to homeostatic conditions after the stressor has dissipated. The cumulative cost of experiencing many stressors and having to undergo allostasis frequently is referred to as "allostatic load," which manifest as the primarily negative health effects of stress.^{1–3} Allostatic load has been associated with chronic diseases including cardiovascular disease, reproductive difficulties, and overall worse health outcomes.^{2,4–6} In addition to diseases and disorders of somatic systems, allostatic load is associated with neuropsychiatric disorders including depression, anxiety, and post-traumatic stress disorder (PTSD). Women are disproportionately affected by the neuropsychiatric repercussions of allostatic load. Women are twice as likely to experience depression in their lifetime compared with their male counterparts.^{7–11} Similarly, the incidence of anxiety and the development of PTSD is higher in women than men.^{12–14} The susceptibility to disease is further increased in women who identify with a minority group and women who have a low socioeconomic status.^{15–19} These epidemiological differences can be attributed to both increased societal pressures on women and sex-specific neurobiological

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responses to experiencing stress.^{6,20} Although societal experiences and pressures can vary between communities, there are established biological differences between men and women that contribute to the differing physical effects of stress on the body, which could contribute to the higher incidence of neuropsychiatric disease in women. This review highlights the existing knowledge and data that explain the physiologic impacts of stress, especially pertaining to neurobiology, and how these impacts differ from sex.

THE HYPOTHALAMIC-PITUITARY-ADRENAL AXIS, STRESS, AND NEUROBIOLOGY

The Hypothalamic-Pituitary-Adrenal Axis

When an organism experiences a psychological stressor, the brain sends signals to prepare the body to respond to a host of potential exposures and experiences, including escape and injury. The mechanism for initiating, sending, and stopping the physical response to stress is achieved through communication between the hypothalamus, pituitary gland, and adrenal gland, often referred to as the hypothalamic-pituitary-adrenal (HPA) axis. When a stressor is perceived, the paraventricular nucleus of the hypothalamus releases corticotrophin-releasing hormone (CRH), which stimulates the anterior pituitary gland in the brain. The anterior pituitary gland then releases adrenocorticotrophic hormone (ACTH), which travels through the bloodstream to stimulate the zona fasciculata layer of the adrenal gland; on stimulation, the adrenal gland increases release of cortisol, which creates physiologic changes throughout the body to effectively respond to the stressor (**Fig. 1**). Cortisol is a steroid hormone that under homeostatic conditions is secreted in a diurnal pattern; thus physiologically, there are higher levels of cortisol in the morning that slowly decrease throughout the day.²¹ Cortisol acts on both mineralocorticoid and glucocorticoid receptors, which are found throughout the body; mineralocorticoid receptors are more involved within homeostasis, whereas glucocorticoid receptors are most involved in stress response activity. Following a stressor, cortisol secretion increases via HPA axis activation, and glucocorticoid receptors serve as the primary effector of stress-induced functions of cortisol. Glucocorticoid receptors are found within the nervous system, immune system, heart, lungs, reproductive and gastrointestinal tracts, muscles, and skin, which allows for a body-wide response to the perceived stressor. On binding to cortisol, the cortisol/glucocorticoid complex acts as a transcription factor to regulate transcriptional changes depending on the particular cell type and previous experiences of the individual. These changes are evolutionarily favored to provide the brain as many nutrients as possible for fuel during stressful events to facilitate the physical ability to generate energy in the form of adenosine triphosphate (ATP) to appropriately respond to a stressor.²² For example, cortisol acts on the liver, muscles, adipose tissues, and pancreas to promote the availability of blood glucose. Readily available blood glucose is critical, as it contributes to the generation of ATP, and thus gives the body more energy for the cardiovascular or muscular system to respond to an imminent threat, such as running away from a fear. The HPA axis is activated when any stressor is perceived whether that be a fear that necessitates escape or a psychological stressor that does not require the same degree of energy mobilization. Thus, psychological stressors can lead to an energy mismatch, as HPA axis-mediated energy mobilization is prioritized without the corresponding physical need for that energy.

Sex Differences in Glucocorticoid Receptors

There are established sex differences associated with the glucocorticoid receptor, which influence the response to a stressor from sex.²³ Glucocorticoid receptors

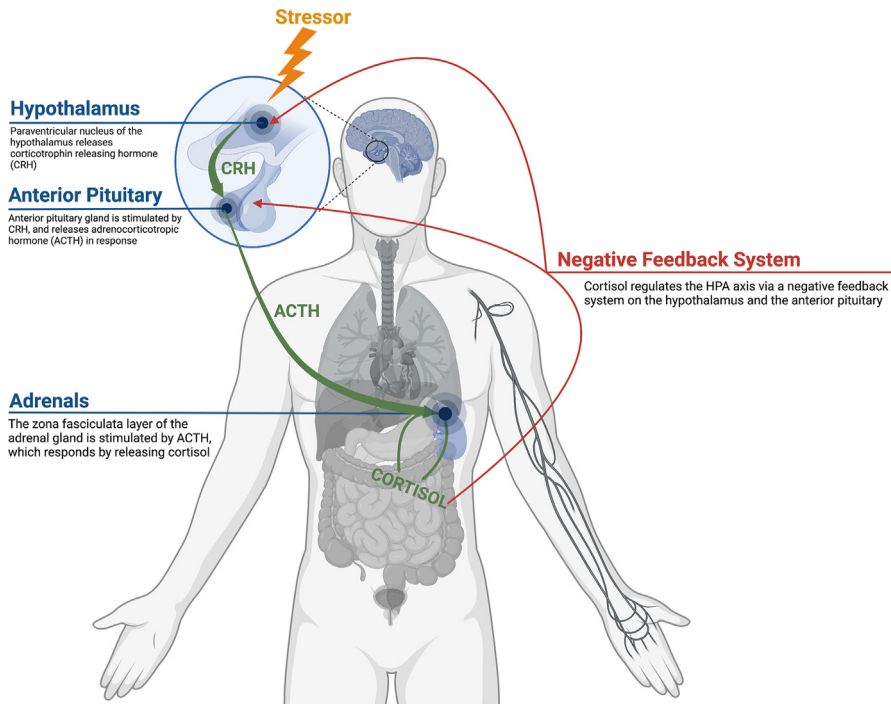


Fig. 1. The HPA axis. Stress activates the paraventricular nucleus of the hypothalamus, which releases corticotrophin-releasing hormone (CRH). CRH stimulates the anterior pituitary gland, which releases adrenocorticotrophic hormone (ACTH). ACTH activates the zona fasciculata layer of the adrenal glands to release cortisol, which travels all throughout the body to coordinate the physical stress response. The HPA axis is turned off via a negative feedback system, when cortisol levels reach threshold, the hypothalamus and anterior pituitary are signaled to turn off, which ceases the physical response to stress. This system becomes faulty during chronic stress, however, as chronic stress impairs this negative feedback loop. (Figure created with [biorender.com](https://www.biorender.com).)

require many steps for activation, which is part of how they are regulated; for activation, cortisol must bind to two glucocorticoid receptors intracellularly, which then form a homodimer and are transported to the nucleus via heat shock proteins and co-chaperone proteins that facilitate translocation, including the proteins PPID and FKBP52, which are both influenced by sex.^{23–25} PPID has been shown to have interactions with estrogen receptors and is transcriptionally upregulated by estradiol.²⁶ FKBP52 has been observed in several breast cancer cell lines, supporting a role of estradiol in its regulation.²⁷ Another protein implicated in the glucocorticoid receptor activation is SRC1, a nuclear coactivator of glucocorticoid receptors, which is also influenced by sex.²³ Testosterone or estradiol supplementation has been shown to increase SRC1 expression, and accordingly SRC1 expression varies during the estrous cycle, illustrating mechanisms by which sex differences and differing hormone levels influence glucocorticoid receptors and thus the HPA axis.^{28–30}

Chronic stress further influences the interaction between sex and glucocorticoid chaperones and associated proteins. For example, chronic stress has been shown to alter glucocorticoid receptor translocation in the form of increased transcription

of the glucocorticoid receptor and co-chaperones such as PPID, FKBP5, and SCR1, which results in attenuation of glucocorticoid receptor translocation and a delayed resolution of plasma corticosterone levels in response to acute stress.³¹ Furthermore, childhood maltreatment is associated with a decrease in methylation of the FKBP5 gene, implicating early-life stress as a mechanism for potentiation of glucocorticoid translocation and alterations in the stress response (p 5).^{32–34} Gene variants in FKBP5 chaperone have also been associated with increased incidence of depression.^{35–37} Thus, stress augments already existing sex differences in the glucocorticoid receptor and corresponding proteins.

HPA Axis and the Brain

The response initiated by HPA axis activation and subsequent engagement of glucocorticoid receptors is also critical for the brain during a stressor, as the brain needs a constant supply of glucose under normal conditions but has an increased demand during times of stress. Facilitative glucose transporters (GLUT), which allow the uptake of glucose into cells, are modified in the brain during perceived stress.^{38,39} At homeostatic conditions, GLUT isoform 3 (GLUT3) is present within the brain, which allows a constant flow of glucose to be supplied to the brain; GLUT3 is specific to the brain and the placenta, because it requires the lowest concentration of glucose to be activated, which prioritizes these structures to receive glucose over other organs in periods of low glucose concentrations.⁴⁰ During times of stress, however, GLUT4, another isoform of these transporters, is upregulated and inserted into neuronal axonal membranes to ensure glucose delivery to the brain in addition to typical delivery via GLUT3.^{38,41,42} GLUT1 is also upregulated in times of cellular stress.³⁹ These changes in GLUT expression in response to stress are location-dependent and chiefly occur in neuroanatomical structures associated with the stress response: the prefrontal cortex (PFC), the amygdala, and the hippocampus.⁴³ The PFC mediates higher order cognition and is activated by HPA axis activation; studies have shown that distinct PFC metabolic glucose patterns are linked to endocrine stress measures as well as subjective perception of stress, indicating the significant role that the PFC has within the stress response.⁴⁴ The amygdala plays a role in HPA axis activation, as it regulates emotional responses and interprets whether something is a stressor warranting HPA axis activation.⁴⁵ The hippocampus is involved in learning and memory and aids in turning off HPA axis activation.^{46,47} The function of each of these areas makes them especially relevant to responding to a stressor, and thus, upregulating GLUT expression in these areas during times of stress is evolutionarily favored. HPA axis activation relies on the brain for perception of a stressor and supports brain function during a stressor in the form of increased energy supply for heightened short-term neurologic functioning.

Chronic Stress, the HPA Axis, and Cortisol

Activation of the HPA axis during stressful events is beneficial, as it is a short-term way to allow efficient physical and neurologic function during stress. However, with chronic stress, the HPA axis becomes dysfunctional. Dysfunction could manifest in an inability to properly turn off or HPA axis activation in response to events that are not actually stressors, such as initiating a response to a perceived stressor that is not proportional to the actual stressor. The HPA axis is regulated via a negative feedback system, when cortisol binds to glucocorticoid receptors in the hypothalamus and pituitary glands, CRH and ACTH secretion is suppressed, which stops the propulsion of the HPA axis response. The hypothalamus is also impacted by neuronal projections from the hippocampus, which when activated further suppress the release of CRH and thus

contributes to halting the stress response.⁴⁶ Under chronic stress or when HPA axis function is compromised, this negative feedback system is impaired, becomes inefficient, and release of cortisol is not regulated appropriately.

The prolonged glucocorticoid exposure that can result from HPA axis dysfunction and exposure to repeated or extreme stressors is associated with an increased risk of developing neuropsychiatric disorders including depression, anxiety, and PTSD.^{6,20} Glucocorticoids are well established as being neurotoxic, especially to areas such as the hippocampus, and thus, maintaining homeostatic levels of cortisol is key.⁴⁸ Acutely, cortisol improves cognition by acting on mineralocorticoid receptors in brain regions specific to the task at hand; for example, learning and memory are improved via stimulation of mineralocorticoid receptors in the limbic system of the brain as that system is highly associated with consolidation and retrieval of memories.^{49,50} A transient increase in cortisol levels can further highlight those pathways associated with learning and memory, hence the benefit of acute HPA axis stimulation. However, when cortisol levels significantly increase, or stay elevated for prolonged periods of time, a disproportionate number of glucocorticoid receptors are stimulated, which conversely decreases cognitive function.^{21,49} Too little cortisol makes completing tasks difficult, but too much cortisol also negatively impacts tasks.^{21,22} For example, study participants who were given exogenous cortisol performed significantly worse in a simple sequence learning task, indicating the neurobiological ramifications of abnormally high levels of cortisol.⁵¹ There is also a correlation between neuropsychiatric diseases such as depression, a disease characterized by cortisol and other neurotransmitter imbalances, and cognition, as patients with major depression performed significantly worse than healthy controls in a battery of tests assessing verbal memory, attention, working memory, and executive function, suggesting imbalanced cortisol as a contributor to these outcomes.⁵² Although just correlative, it is reasonable that imbalanced cortisol could be mediating negative cognitive outcomes considering that the addition of exogenous cortisol resulted in similar findings. This is further supported by findings from patients who require long-term corticosteroid medication to manage a particular disease, for example, patients on steroids to prevent multiple sclerosis flare ups; long-term corticosteroid use has been found to increase neuropsychiatric disorders such as depression, increase mood fluctuations, and decrease cognitive function.^{53–59} Glucocorticoids have a negative impact on memory, attention, mood, and cognition, suggesting a connection between chronic stress and increased incidence of neuropsychiatric diseases.

Furthermore, there are sex-specific effects of cortisol imbalances on neurocognition. For example, low-dose hydrocortisone administration transiently enhances cognition in women, a finding that was not consistent in men; this effect is present up to 4 hours of dosing but fades and becomes detrimental beyond 4 hours of hydrocortisone exposure.⁶⁰ This further indicates how HPA axis activation can be initially beneficial but can quickly become damaging with over-activation from chronic stress. In addition, women who were treated with low-dose hydrocortisone in order to explore cortisol as a mechanism for neurologic changes had a heightened ability to recognize fearful faces when compared with men; this heightened efficacy in threat detection was only present in women, and especially women who had lower psychological burdens and who had experienced minimal trauma. This enhanced threat detection was lost in women who had experienced a significant amount of trauma, indicating that cortisol can be beneficial for responding to stressful situations, but this system becomes dysfunctional in women who experience settings of prolonged stress exposure and long-term increases in glucocorticoids.⁶¹ In another study, high cortisol was again associated with worse memory and visual perception in both women and men, but in

individuals with high cortisol, a variety of microstructural brain changes were seen in women only.⁶² Similarly, cortisol levels that are too low also negatively impact cognition, as there are not high enough amounts to activate mineralocorticoid receptors.⁵⁰ The impact of cortisol on learning is also influenced by past experiences specific to individuals, indicating how life experiences and dispositional traits can influence these pathways.^{49,63–65} Potential mechanisms of these individualistic influences have been hypothesized to be of an epigenetic origin of the glucocorticoid receptor and its expression, but more research is warranted to fully understand how individual experiences play a role on resulting stress responses.⁶⁶

The impact of chronic stress on the brain can endure long after the stressor exposure has concluded, and these effects are particularly pronounced in women as compared with men. For example, imbalanced cortisol, indicative of HPA axis dysregulation following chronic stress, can persist in some female patients who have experienced a depressive episode in the past, regardless of whether that depression is in remission or active; this indicates the possibility of lasting effects of chronic stress on physiologic processes for women, as this pattern is not present for men.⁶⁷ Furthermore, patients with remitted major depressive disorder demonstrated a decrease in learning and memory, suggesting that there are lasting impacts of neuropsychiatric diseases.⁶⁷ However, more research is warranted to understand those dynamics and different mechanisms contributing to the effects of stress, as that is not the case for all women who have experienced a depressive episode. Lasting effects of stress on structural and functional neurologic differences between men and women are heightened by imbalanced cortisol levels, and women seem to be especially susceptible to potential changes.

The negative impacts of stress and neuropsychiatric disease on cognitive function co-occur with neuroanatomical modifications. Childhood trauma has been shown to correlate negatively with brain volume, indicating the physical effect stress can have on neuroanatomy.⁶⁸ Patients who have experienced chronic stress and depression have also been found to have decreased amounts of gray matter; patients' hippocampus, amygdala, and PFCs were all affected. Depression has been associated with structural and functional alterations in the hippocampus and medial PFC.⁶⁹ Specifically, structural imaging of the hippocampus in patients diagnosed with major depressive disorder showed that these patients consistently had reduced hippocampal volumes when compared with healthy controls. Patients who had other neuropsychiatric conditions beyond depression had even more significant reductions in hippocampal and PFC volumes, indicating a more significant effect with additional neuropsychiatric conditions.^{70–74} Neuroanatomical changes resulting from stress are also influenced by sex. In a study, the association between early morning serum cortisol, cognitive performance and brain structural integrity, and higher cortisol was inversely associated with cerebral brain volume in women; high cortisol levels resulted in decreased regional fractional anisotropy, a measure of connectivity within the brain, especially in the splenium of corpus callosum and the posterior corona radiata, two areas critical to coordination between brain hemispheres and neural traffic.⁶² The experience of trauma as a form of chronic stress has also been shown to alter the neuroanatomy of women. In a study of mothers who had experienced childhood maltreatment and those who had not, maltreatment was associated with greater gray matter volume in brain regions associated with emotional empathy and less maternal sensitivity when interacting with their children.^{75,76} Although these studies and the effects of stress and sex on neuroanatomical changes are still being investigated, assessing neuroanatomical changes as a basis of stress exposure and sex is an important consideration.^{77–80} These studies all demonstrate that dysfunctional

HPA axis activation by chronic stress has physical ramifications and that women experience different neurobiological changes than men as a result of stress (**Fig. 2**).

THE HYPOTHALAMIC–PITUITARY–GONADAL AXIS

Hypothalamic–Pituitary–Gonadal Axis

One explanation for sex-specific differences in neurobiological responses to stress is differing hormone profiles between men and women, the system of which is referred to as the hypothalamic–pituitary–gonadal (HPG) axis. Similar to the HPA axis, where the brain perceives a stressor, the hypothalamus secretes CRH to stimulate the pituitary to secrete ACTH to stimulate the adrenals to secrete glucocorticoids, the HPG axis is characterized by hypothalamus and pituitary influence over the gonads and reproductive tract.⁸¹ The parvocellular neurosecretory cells within the hypothalamus secrete gonadotropin-releasing hormone, which stimulates the anterior pituitary.⁸¹ On stimulation, the anterior pituitary releases luteinizing hormone (LH) and follicle-stimulating hormone (FSH) into the general circulation, which act on the gonads of both men and women. In response to LH and FSH, the testes primarily release testosterone, an androgen, and the ovaries release estradiol, an estrogen. The HPG axis becomes activated around 10 weeks of gestational development and, in conjunction with sex chromosomes, drives organizational, or permanent, sex differences in males in females.⁸² It continues throughout the lifespan to influence activational or transient sex differences via these different hormone profiles.⁸²

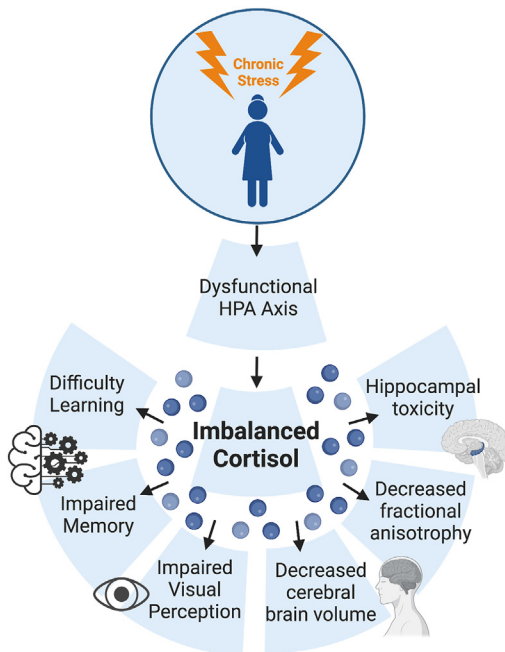


Fig. 2. Lasting neurobiological effects in women with a history of chronic stress. Chronic stress is associated with a variety of lasting physical effects in women. It results in a dysfunctional HPA axis, which leads to imbalanced cortisol levels even after the stress is no longer present. Imbalanced cortisol mediates neurobiological changes in women, such as cognitive dysfunction and neuroanatomical modifications. (Figure created with [biorender.com](https://www.biorender.com).)

Estrogen

One of the significant hormones that potentiate sex-differences via the HPG axis is estrogen. There are three predominant estrogens—estrone (E1) is a weak estrogen with higher levels present after menopause.⁸³ Estradiol (E2) is predominant throughout the lifespan and the most common form of estrogen, and estriol (E3) is a weak estrogen that increases during pregnancy, as it is important for maintaining the fetus.^{84,85} Both males and females are physically affected by estrogen, but the degree of which and the location of these effects are part of what distinguishes sex. During development, when the HPG axis is first initiated, estrogen receptors and estrogen signaling in the brain have a significant impact on brain development; in males, the testes briefly activate at birth, give a surge of testosterone, which is converted to estrogen via neural aromatase.⁸⁶ This estrogen works via estrogen receptor alpha to establish sex-specific differences in cell number and connectivity, an organizational effect of estrogen, which lays the groundwork for neurobiological differences between sexes in conjunction with sex chromosomes.^{84,87,88} Following development, estrogen continues to have physical influences specific to sex by influencing protein expression and function via activational effects. First, the degree of which estrogen is produced varies between men and women and varies as a function of age. Men produce primarily testosterone from the testes in a consistent fashion and experience some peripheral conversion to estrogen and progesterone, whereas women produce primarily estrogen and progesterone from the ovaries in a cyclical pattern, dependent on age, with a small amount of testosterone produced peripherally.⁸⁹

Estrogen has been implicated as a key regulator of neurotransmitters and correlative neurotransmitter circuitry. Estrogen influences transcriptional changes via nuclear receptors, specifically estrogen receptors alpha and beta, which on binding serve as a transcription factor complex and directly influence gene expression in neural cells.⁹⁰ For example, estrogen has been found to increase dopamine synthesis in the brain and regulate dopamine receptors in response to varying dopamine levels.^{91–94} It also upregulates the expression of tryptophan hydroxylase, an enzyme involved in synthesizing serotonin, to increase serotonin biosynthesis in the brain.^{91,92,95} In addition, estrogen regulates the serotonin receptors 5HT₂ and 2C in response to varying serotonin levels.^{96,97} The regulation of dopamine and serotonin allows estrogen to be a chief regulator of mood and behavioral responses and thus is further implicated in neuropsychiatric disorders that so frequently modulate mood and behavioral responses. Estrogen as a transcription factor also has benefits to cognitive function, especially learning and memory, as it has been shown to upregulate N-methyl-D-aspartate (NMDA) glutamate receptors, which are critical to learning and memory.^{91,98–100} Considering the detrimental impact of neuropsychiatric diseases on learning and memory, it is noteworthy that estrogen has a role in both.

Beyond influencing genomic effects via transcription factors, there are also membrane-bound estrogen receptors alpha and beta as well as membrane bound G-protein-coupled estrogen receptors, which can facilitate non-genomic effects, further expanding estrogen's influence in the brain.¹⁰¹ Membrane-bound estrogen receptors facilitate rapid responses resulting from estrogen, rather than pathways characterized by nuclear receptor activation which typically require time to facilitate transcriptional changes.¹⁰¹ Rapid estrogen signaling via these receptors can lead to axospinous synapse formation in the hippocampus and increases structural plasticity of postsynaptic neuronal spines, which both enhance glutamatergic transmission and long-term potentiation and plasticity, further supporting estrogen's role in learning and memory.¹⁰¹ These different receptors complement the transcriptional function of

estrogen, and the combination of both makes estrogen a key modulator of neurotransmitter systems and learning and memory within the brain.

Because estrogen and other HPG axis sex hormones influence transcription and work via nuclear receptors, similar to the glucocorticoid receptor and proteins associated with the HPA axis, there are key interactions between the HPG and HPA axes, as there are overlapping proteins and chaperones involved within each. Thus, the HPG and HPA axes can transactivate and transrepress each other. For example, FKBP5, one of the co-chaperones that impact glucocorticoid translocation to the nucleus and thus the potentiation of the HPA axis-mediated stress response influences and is influenced by estrogen.¹⁰² FKBP5 has been found to interact with nuclear receptors beyond the glucocorticoid receptor, including estrogen nuclear receptors.¹⁰² Estrogen has been found to potentiate the increase in FKBP5 expression in conjunction with cortisol.¹⁰³ Conversely, estradiol treatment resulted in decreased expression of peptidylprolyl isomerase D (PPID), another key glucocorticoid receptor chaperone, an effect that was further exacerbated by the administration of progesterone.¹⁰³

Estrogen specifically has been shown as a key regulator within the HPA axis in both men and women.¹⁰⁴ For example, estrogen influences CRH neurons which are key regulators within the limbic system of sensory inputs and play a significant role in identification of physiologic stressors and thus the HPA axis; these neurons contribute to regulation of behavior, perception, and autonomic and neuroendocrine responses.^{104–106} Estrogen receptor alpha is expressed throughout CRH neuronal circuitry, and treatment with estrogen has been shown to increase CRH neuronal activity, thus increasing CRH levels, and furthering the HPA axis response.^{107,108} In addition, in a preclinical study assessing the impact of chronic stress in offspring from mothers exposed to chronic prenatal stress, estrogen was found to mediate an increase in visceral hypersensitivity as a result of chronic stress in these offspring via an increase in brain-derived neurotrophic factor expression, potentiating the HPA axis response to chronic stress.¹⁰⁹ Estrogen can also work against the HPA axis and mitigate the physical stress response. For example, high concentrations of estrogen receptor beta have been identified within neurons within the paraventricular hypothalamic nucleus, which are classically antagonistic to CRH-releasing neurons, and thus mitigate HPA axis activation.¹¹⁰ Furthermore, estrogen can work in a synergistic manner to oxytocin, a hormone associated with maternal behaviors, parturition, lactation, and mitigating the physical stress response launched by the HPA axis. Estrogen enhances oxytocin's effect via modulation of oxytocin neuron function via estrogen receptor beta-driven reductions of cortisol and ACTH levels within these hypothalamic paraventricular nucleus (PVN) neurons.^{104,111}

These differing effects of estrogen within HPA axis regulation are likely due to differences in estrogen receptor alpha and estrogen receptor beta. Estrogen receptor alpha has been found to facilitate estrogen-mediated impairment of the negative feedback loop on the HPA axis.^{112,113} In a study assessing estradiol treatment on the effects of the synthetic glucocorticoid, dexamethasone, treatment with an estrogen receptor alpha agonist increased stress-induced cortisol via activation of gamma-aminobutyric acid (GABAergic) neurons; conversely, treatment with an estrogen receptor beta agonist resulted in the opposite effect, primarily via the reduction of glucocorticoid receptors.¹¹² In another study, peripheral treatment with an estrogen receptor beta agonist prevented HPA axis reactivity as well as decreased anxiety-like behaviors seen in female rats in response to a glucocorticoid receptor agonist.¹¹⁴ Both of these findings suggest a strong role of estrogen within HPA axis regulation, either in upregulation of the stress response via estrogen receptor alpha or downregulation via estrogen receptor beta. The ability of estrogen to increase HPA axis hyperactivity via

disruption of the negative feedback loop can potentiate increased cortisol in women who experience chronic stress. Furthermore, the neuroprotective aspect of estrogen via activation of estrogen receptor beta could explain why neurobiological effects of childhood trauma can be shielded until menopause, a time when estrogen levels are decreased significantly in women.¹¹⁵

More research is warranted to discern the balance between the neuroprotective effects of estrogen and the transcriptional changes that can increase stress susceptibility mediated by estrogen as well as estrogen's conflicting role in HPA axis hyperactivity. Regardless, estrogen and similar sex hormones provide a significant basis for neurobiological changes resulting from stress as a function of sex and contribute to the increased physical burden of stress on women. The increased concentrations of circulating estrogen within women, as well as the fluctuations women experience throughout their lifespan, heighten these effects for women, which is better understood by assessing the incidence and resulting impacts of stress and neuropsychiatric disease in women throughout the lifespan.

ESTROGEN THROUGH THE LIFESPAN, NEUROPSYCHIATRIC DISEASE, AND CLINICAL INTERVENTIONS

Estrogen and Stress Through the Lifespan

Estrogen and other sex hormones change throughout a woman's life with the core hormonal stages being prepubertal, puberty and reproductive years, perimenopause, and menopause.^{116–118} Prepubertal males and females have similar hormone levels with the biggest differences developing during puberty. That being said, prepubertal children are still susceptible to neurobiological changes due to stress. For example, early abuse and neglect have been found to reduce resilience to stressful events and decreased ability to emotionally regulate responses.¹¹⁹ Neuroimaging studies in individuals who experienced childhood trauma indicate hypoactivity in the frontal lobe and corresponding decreases in executive function.¹¹⁹ Childhood trauma has been associated with hypersensitivity to threat and signs of anger and a corresponding increase in synapse formation and dendritic spines in the amygdala, where threats and emotions are processed.¹¹⁹ Children who have experienced childhood trauma have been found to be more susceptible to having difficulty regulating their emotions.¹¹⁹ Furthermore, there is an association between experiencing childhood trauma and the development of psychosis with the number of traumatic events increasing the likelihood of developing psychosis.^{120,121} Early life stress has been associated with complex clinical profiles of both neurobiological changes and other diseases.^{122–124} Although there is no known sex difference in the experience of early life stress in the prepubertal phase, experiencing early life stress correlates to neuropsychiatric diseases more significantly in women later in life.¹²⁵

There are two stages of puberty: adrenarche, hallmarked by adrenal androgen production and gonadarche, hallmarked by the activation of the HPG axis.^{126,127} During puberty, males have an increase in testosterone while females undergo menarche, the initiation of the menstrual cycle, which introduces another layer of hormonal fluctuations beyond what is seen within different life stages.¹²⁸ During each menstrual cycle, estradiol levels fluctuate to prepare the reproductive system for fertilization and shed the uterine lining that is created if fertilization does not occur. The average menstrual cycle is 28 days. From days 0 to 12 estradiol gradually increases, and it sharply decreases on ovulation, from days 13 to 17. After ovulation, estradiol increases again from day 18 to 24 and decreases again from day 24 to 28, which is when uterine shedding occurs.¹²⁹ Neurobiology is affected by these monthly fluctuations of estradiol.

Structural and functional neural plasticity increases throughout the menstrual cycle as estradiol levels vary, and mood changes are more likely to occur during the period of low estradiol that precedes ovulation.^{130–132} With the onset of puberty, women enter their reproductive years, where they are twice as likely to develop depression when compared with their male counterparts.^{10,20,133} Furthermore, the timing of trauma during puberty can influence the development of neuropsychiatric diseases: in a study of pubescent girls, trauma during puberty conferred significantly more risk of anxiety disorder diagnoses, whereas prepubertal trauma conferred significantly more risk of depressive disorder diagnoses, indicating that timing of trauma and stress can influence resulting neuropsychiatric outcomes.¹³⁴ Early life stress can also affect the timing of puberty. Experiencing early life stress is positively associated with the development of early puberty in girls and late puberty in boys when compared with their peers.¹³⁵ Early life puberty was predicative of the development of anxiety in girls but not boys, indicating further differences between early life stress and how stress impacts neuropsychiatric outcomes.¹³⁵

The role of estrogen in neurobiological changes as a result of stress becomes even more apparent when considering incidence of neuropsychiatric diseases within women during perimenopause and after. Increased susceptibility to developing depression is further heightened when women enter perimenopause, the period before menopause when periods become irregular and estrogen levels begin to decrease.¹³⁶ On average, perimenopause begins in the late 40s and lasts 4 to 5 years until women stop having a menstrual period completely and they enter menopause.¹³⁶ The more significant hormone fluctuations introduced in the perimenopausal stage, and the subsequent decreases in estrogen could contribute to the higher incidence of neuropsychiatric diseases in this population. Furthermore, it has been well established that childhood trauma or experiencing chronic adolescent stress is associated with the development of neuropsychiatric disorders and physical ramifications in both males and females.^{119,124} That being said, the increase in incidence of neuropsychiatric disorders in women during the perimenopausal phase has been associated with experiencing childhood trauma—a woman is more likely to develop depression during perimenopause if she experienced childhood trauma, a pattern that is not observed for men.¹³⁷ In addition, the effects of childhood trauma can be shielded until menopause, when neuroprotective estrogen is depleted.^{137,138} Furthermore, estrogen treatment has shown to be effective as an antidepressant in women who experience depression during perimenopause, further implicating hormonal changes as a chief modulator for neurobiological changes and the potential development of psychiatric diseases.^{136,139} Because of the increased susceptibility to developing depression in this population, women undergoing perimenopause and then menopause should be consistently screened for depression.¹³⁸

These differences in neurobiological changes as a result of stress as a function of age and pubertal stage implicates estrogen as a key modulator. Understanding neuroprotective and deleterious effects of estrogen is essential to further elucidating these dynamics, especially considering the effect of cycling gonadal steroids on the HPA axis and neurotransmitter systems in women.¹⁴⁰

Clinical Interventions

Considering the significant physical consequences that women especially experience as a result of chronic stress, it should be evident how beneficial interventions mitigating stress would be. The neurobiological changes that women especially experience as a result of stress have implications for disease; HPA axis dysregulation has been associated with not only depression but also cardiovascular disease, diabetes,

hypertension, suppressed immune function, and impaired wound healing.^{140,141} Accordingly, interventions aimed at mitigating stress should be taken as seriously as preventative interventions to mitigate diabetes, as the physical effects are similarly impactful. Although there are some interventions that seem efficacious in reducing stress and remission rates in people who experience neuropsychiatric diseases, more research should be done to investigate different interventional approaches. For example, eye movement desensitization and reprocessing and trauma-focused cognitive behavioral therapies have both been found to decrease remission rates and symptoms associated with PTSD.^{142,143} Simple mindfulness interventions have also been found to decrease stress in a variety of populations.^{144–149} Interventions have various levels of involvement and detail, and yet can still have similar preventative effects, which exemplifies how these interventions can be easily accessible while still having significant potential to decrease these significant physical changes that result from stress.

SUMMARY

This review provides an overview of neurobiological differences between men and women, and how men and women have alternate responses to stress as a result of these neurobiological changes. These conclusions should hopefully impact medical practitioners, policy makers, and researchers to understand the following:

Practice: Awareness that stress, regardless of if that is an established clinical disease such as depression, PTSD, anxiety, or if that is just a general stressful state, impacts women via direct physical ramifications rather than being a fleeting emotion.

Policy: Interventions aimed at mitigating the burden of chronic stress should be highly considered, especially in light of all of the physical processes associated with chronic stress.

Research: More research is warranted to further elucidate the mechanisms underlying the increased susceptibility to physical changes women experience when compared with men as a result of stress. Further investigation of the differing effects of stress on estrogen and correlative enzymes such as aromatase should especially be considered.

CLINICS CARE POINTS

- Women are significantly more likely to develop a neuropsychiatric disorder when compared with their male counterparts, a major risk factor of which is experiencing chronic stress.
- Chronic stress can lead to HPA axis dysfunction in women.
- Women experience long-term imbalanced cortisol and resulting impacts as a result of chronic stress, whereas men are not as longitudinally impacted.
- Estrogen levels are a key protein difference between women and men that could be contributing to the different neurobiological impacts of chronic stress.
- Interventions aimed at mitigating chronic stress are warranted especially in women, considering the long-term physical effects stress has on the body.

DISCLOSURE

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